Algorithmic Trading and Firm Value

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Abstract

Using data from 2002 to 2013, we examine the impact of algorithmic trading on firm value. The results show that algorithmic trading generates net benefits for firm value through impacting stock liquidity, idiosyncratic volatility, and idiosyncratic skewness, and firms benefit more from algorithmic trading when algorithmic trading intensity is higher. Using the advent of auto quotation on the New York Stock Exchange as an exogenous shock to algorithmic trading, we find evidence of a causal effect of algorithmic trading on firm value through channels of stock liquidity, idiosyncratic volatility, and idiosyncratic skewness. The positive effects of algorithmic trading on firm value are stronger for firms with higher stock liquidity, for larger firms, and in the post-2007 period when algorithmic trading intensity is higher.

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1. Introduction

As commonly used, algorithmic trading refers to the use of computer algorithms to make trading decisions, submit orders, and manage orders after submission. It is a dramatic example of technological change that revolutionized the trading of securities. Beginning in the U.S. equity market in the late 1990s, the use of algorithmic trading has become more common and predominant in major financial markets, representing an estimated 78% of all U.S. equity trading volume in 2012.[1] The literature has documented that improvements in market microstructure due to technological advances can have a strong impact on firm fundamentals. For example, Sanger and McConnell (1986) examine the increased transparency and liquidity provided to OTC stocks with the creation of Nasdaq in 1971 and find results that suggest an increase in the value of OTC firms. Amihud, Mendelson and Lauterbach (1997) show that stocks that were transferred to a more efficient trading mechanism in the Tel Aviv Stock Exchange in 1987 enjoyed significant and permanent price increases. Easley, Hendershott and Ramadorai (2014) find that the Phase II of the 1980 upgrades to the NYSE's trading environment had a positive effect on liquidity which resulted in positive abnormal returns for those stocks that received the upgrade. Given the significant role of algorithmic trading in the stock market, investors, academics and policymakers need a deeper understanding of its benefits, costs, and real effect, including those on firm fundamentals.

The extant literature on algorithmic trading has mainly focused on its impact on market quality, but its effect on firm fundamentals and firm value, which is our focus, remains unknown. Some extant studies conclude that algorithmic trading benefits capital markets by increasing trading volume and liquidity, reducing trading costs, contributing to price discovery and increasing market

[1] See http://neverlosstrading.com/Algorithmic%20Trading.html. According to Credit Suisse, which cited data from TABB Group, high frequency trading, the major subset of algorithmic trading, accounted for about 50% of all US equity trading in 2016 (https://www.marketwatch.com/story/high-frequency-trading-has-reshaped-wall-street-in-its-image-2017-03-15).

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efficiency (see, e.g., Brogaard, 2011; Brogaard, Hendershott, and Riordan, 2014; Frino et al., 2017), while others conclude it increases volatility, reduces price informativeness and harms capital markets (see, e.g., Zhang, 2010; Scholtus, van Dijk, and Frijns, 2014; Boehmer, Fong, and Wu 2018; Weller, 2018; Breckenfelder, 2019). Moreover, algorithmic trading, especially the major subset of it known as high frequency trading, has been criticized for contributing to the 2010 Flash Crash[2] (e.g., Kirilenko et al., 2017). Collectively, the literature provides mixed results on the question of whether algorithmic trading is a beneficial financial innovation, leaving it the subject of intense public debate and controversy.

We contribute evidence to the debate by examining the impact of algorithmic trading on firm value. If algorithmic trading increases liquidity, as shown in Hendershott, Jones, and Menkveld (2011), it should improve the information content of stock prices and reduce information asymmetry and cost of capital, leading to higher firm value. Improved stock price information content could also increase the effectiveness of managerial compensation contracts that are based on stock prices and improve managerial investment decisions (Fang, Noe, and Tice, 2009). In contrast, if algorithmic trading reduces liquidity, it may increase equity risk and cost of capital, resulting in lower firm value. Algorithmic trading may also affect firm value through other channels. Stock return volatility, especially, idiosyncratic volatility, is positively related to expected stock return as predicted by theories (e.g., Merton, 1987) and confirmed empirically (e.g., Fu, 2009). Under-diversified investors demand a higher return as a compensation for bearing higher idiosyncratic volatility. Some investors prefer stocks with positive skewness and place a higher value on them, resulting in a negative relation between idiosyncratic skewness and expected return (Mitton and Vorkink, 2007; Boyer, Mitton and Vorkink, 2010). Lower expected return indicates higher current

[2] On May 6, 2010, the Dow Jones Industrial Average plunged about 1000 points, the second largest intraday point swing ever to that date and is commonly referred to as the "Flash Crash".

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stock price and firm value. The existing literature suggests that algorithmic trading may have differing effects on stock characteristics such as liquidity, volatility, and price efficiency, and its short-term and long-term impacts may also differ.[3] These potentially contradicting complexities make it difficult to assess the overall impact of algorithmic trading on financial markets and individual securities. By examining the effect of algorithmic trading on firm value, we hope to augment extant studies on the effects of algorithmic trading and provide evidence useful for evaluating the net overall costs and benefits of algorithmic trading.

To examine the relation between algorithmic trading intensity and firm value, we follow Hendershott, Jones, and Menkveld (2011) and Boehmer, Fong, and Wu (2018) to construct an algorithmic trading proxy based on message traffic using the Trades and Quotes (TAQ) database. We first examine how algorithmic trading affects various stock price charateristics, specifically, stock liquidity, idiosyncratic volatility, and idiosyncratic skewness, and find that algorithmic trading is significantly positively related to stock liquidity and idiosyncratic skewness and significantly negatively related to idiosyncratic volatility. We then find that algorithmic trading activity is significantly positively related to firm value measured by Tobin's q after controlling for other variables that affect firm value. The relation is robust to the inclusion of year, industry, and firm-fixed effects, different model specifications, and the use of alternative proxies for algorithmic trading activity. These results suggest that algorithmic trading positively impacts firm value through improving stock liquidity, decreasing idiosyncratic volatility, and increasing idiosyncratic skewness.

To help establish a causal effect of algorithmic trading on firm value and address endogeneity concerns, we use the change to automated quote dissemination (auto quote) by the New

[3] See for example, Boehmer, Fong, and Wu (2018), Hendershott and Riordan (2013), Brogaard, Hendershott, and Riordan (2014), Brogaard (2011), and Zhang (2010).

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York Stock Exchange (NYSE) in 2003 as an exogenous shock to algorithmic trading intensity of NYSE stocks. Auto quote provides quicker feedback to algorithms and results in more electronic message traffic and higher algorithmic trading intensity. We estimate two-stage least squares regressions on NYSE stocks over a narrow window of two quarters surrounding the implementation of auto quote. The results provide evidence for a causal effect of algorithmic trading on firm value. In addition, we find results suggesting that parts of the positive effect of algorithmic trading on firm value are through the channels of stock liquidity, idiosyncratic volatility, and idiosyncratic skewness. The three channels altogether account for about two thirds of the effect of algorithmic trading on firm value. Among them, the liquidity channel is the main mechanism, followed by the idiosyncratic volatility and skewness channels.

We further examine how the effect of algorithmic trading on liquidity and firm value depends on stock and firm characteristics and time periods. Algorithmic trading intensity varies across different groups of stocks and time periods, and it may have a nonlinear impact on firm value. Recent literature shows that large and liquid stocks that are subject to higher intensity of algorithmic trading benefit more from algorithmic trading in terms of improved liquidity (Hendershott, Jones, and Menkveld, 2011; Boehmer, Fong, and Wu, 2018). Consistent with those findings, we find that the positive effects of algorithmic trading on firm value are significantly stronger for more liquid stocks, larger firms, and after 2007 when the overall level of algorithmic trading intensity is higher.

Our study is not designed to rule out the existence of any harmful effects of algorithmic trading, so we cannot and do not conclude that it has no harmful effects. It is conceivable that algorithmic trading leads to negative effects such as flash crashes that manifest themselves periodically while still providing net benefits to firm value for stocks with greater algorithmic

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trading. On the narrow, but important, firm value dimension, however, we can conclude that algorithmic trading is associated with increased stock liquidity, decreased idiosyncratic volatility, increased idiosyncratic skewness and increased firm value, especially when algorithmic trading intensity is high.

We contribute to the literature on algorithmic trading and high frequency trading in a unique way by directly testing the relation between algorithmic trading activity and firm value. Prior studies on algorithmic trading and high frequency trading focus on their impacts on measures of market quality, but little is known about the effect of algorithmic trading on firm fundamentals and ultimately on firm value.[4] Our results on firm value, to our knowledge the first such evidence, are critical in assessing the net social benefits of algorithmic trading. In addition, our study contributes to the algorithmic trading literature by providing new evidence that algorithmic trading reduces idiosyncratic volatility and increases idiosyncratic skewness.

Our study also contributes to the strand of literature at the intersection of market microstructure and corporate finance or asset pricing. Financial markets can reflect firm fundamentals, and they can also *affect* firm fundamentals. Baker, Stein, and Wurgler (2003), Luo (2005), and Chen, Goldstein, and Jiang (2007) provide evidence that market prices affect firms' investments via managerial learning and/or the firm's access to new capital. Roll, Schwartz, and Subrahmanyam (2009) find firms with more options trading have higher values of Tobin's q. Fang, Noe, and Tice (2009) find that greater market liquidity increases the information content of market prices and of performance-sensitive managerial compensation, and thereby positively impacts firm value. The link between the stock characteristics and asset prices are typically established using standard asset pricing tests that ignore causality and endogeneity. Our study contributes to this line of literature by showing that algorithmic trading affects firm value through impacting the stock

[4] Boehmer, Fong, and Wu (2012) study the effect of algorithmic trading on firms' equity raising.

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characteristics of liquidity, volatility and skewness. Using exogenous variations in algorithmic trading due to the implementation of the auto quote as instruments, our study provides evidence for causal effects of liquidity, idiosyncratic volatility and idiosyncratic skewness on firm value.

The rest of the paper proceeds as follows. Section 2 reviews the literature and develops our hypotheses. Section 3 describes the sample, data sources and variable measurements. Section 4 presents the main empirical results, and Section 5 concludes.

2. Literature Review and Hypotheses

Due to advances in technology and the ability to trade at much higher frequencies, financial markets and trading have changed in fundamental ways (Kirilenko and Lo, 2013; O'Hara, 2015). Given the predominance of algorithmic trading and its controversial role in equity markets, a central question that interests financial market participants, policymakers, and academics is whether algorithmic trading should be regulated and scrutinized more intensely. The algorithmic trading literature has focused on the impact of algorithmic trading on market quality and provides mixed evidence. On the one hand, a line of literature finds a positive impact of algorithmic (or high frequency) trading on market quality. Brogaard (2011) finds that high frequency trading reduces intraday volatility in the short-term, and Brogaard, Hendershott, and Riordan (2014) find that overall high frequency trading enhances price efficiency. Frino, Prodromou, Wang, Westerholm, and Zheng (2017) show that algorithmic trading accelerates the information incorporation process and improves market efficiency. Similar evidence has also been found for algorithmic trading in other financial markets. Viljoen, Westerholm, and Zheng (2014) find evidence that algorithmic traders contribute to liquidity and price discovery on the futures market. Chaboud, Chiquoine, Hjalmarsson, and Vega (2014) provide evidence that algorithmic trading in foreign exchange

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markets contributes to a more efficient price discovery process and has a positive impact on market liquidity without creating excess volatility.

On the other hand, several studies suggest that algorithmic trading or high frequency trading creates excess volatility and may be detrimental to financial markets. For example, Kirilenko, Kyle, Samadi, and Tuzun (2017) study how high frequency trading is related to the May 6, 2010 Flash Crash and conclude that although high frequency trading did not cause the crash, it worsened the crash by exacerbating market volatility. Zhang (2010) finds that high frequency trading is positively correlated with stock price volatility, and negatively related to the market's ability to incorporate information about firm fundamentals into asset prices. Scholtus, van Dijk, and Frijns (2014) find that algorithmic trading around macroeconomic news increases volatility and reduces overall depth. Boehmer, Fong, and Wu (2018) conduct an international study using message counts as a proxy for algorithmic trading and show that algorithmic trading increases volatility. Weller (2018) hypothesizes that although algorithmic trading facilitates incorporating existing information into stock prices, improving price efficiency, it simultaneously discourages acquiring new information and reduces available information and stock price informativeness. Consistent with that hypothesis, he finds that algorithmic trading reduces stock price informativeness. Breckenfelder (2019) finds that competition among high-frequency traders reduces market liquidity and increases volatility.

Many studies examine the impact of algorithmic trading on liquidity. Using NYSE electronic message traffic as a proxy for algorithmic trading activity, Hendershott, Jones, and Menkveld (2011) find that algorithmic trading improves liquidity and enhances the informativeness of quotes. Boehmer, Fong, and Wu (2018) provide international evidence showing similar results. Using a sample of algorithmic trading on the Deutsche Bourse, Hendershott and Riordan (2013) show that algorithmic traders consume liquidity when it is cheap and provide liquidity when it is

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expensive, which reduces liquidity risk. Conrad, Wahal, and Xiang (2014) use data from 2009 to 2011 and find that higher quote activity (any change in the BBO price or size) in US stocks is associated with greater price efficiency and lower trading costs. They also find that an exogenous shock to the Tokyo Stock Exchange that reduces latency and allows co-location (and thus enhances the environment for algorithmic trading) leads to improved price efficiency and a decline in trading cost. Groth (2011) finds that algorithmic trading does not create excess volatility and does not reduce liquidity during periods of high volatility. And finally, looking at pre-algorithmic trading, Easley, Hendershott, and Ramadorai (2014) find that a technology change on the floor of the NYSE in 1980 improved liquidity which led to an increase in share prices.

Consistent with some of the complaints from market participants, there are also studies that suggest that increases in algorithmic trading can harm liquidity. Using a measure of abnormal standardized quoting activity as a proxy for algorithmic trading, Upson and Van Ness (2017) find that in the first quarter of 2012 that execution quality decreases with increases in algorithmic trading. Chakrabarty, Jain, Shkilko, and Sokolov (2018) examine changes in quoting activity around the implementation of the naked access ban. They find a decrease in quoting activity that leads to a decline in effective spreads, quoted spreads, and an increase in depths at the inside quotes. They interpret this as evidence that when certain types of algorithmic quoting activity are eliminated, liquidity costs improve. Lyle and Naughton (2016) provide evidence that increases in algorithmic trading prior to 2007 lead to lower trading costs while subsequent increases in algorithmic trading, which were not accompanied by increases in market maker monitoring, have not improved market quality.

Liquidity can be linked to firm value in two ways. First, both theoretical and empirical asset pricing studies have indicated that liquidity is priced by the market. Amihud and Mendelson (1986)

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model effects of bid-ask spreads on asset returns and predict that expected return is positively related to illiquidity. O'Hara (2003) argues that asset pricing models should incorporate transaction costs of liquidity. Acharya and Pedersen (2005) derive a liquidity-based capital asset pricing model and show that an increase in liquidity results in higher contemporaneous returns and lower expected future returns. Thus, greater liquidity should lead to higher current stock prices and greater firm value. Most of the empirical studies confirm the pricing of liquidity, although the evidence is sometimes complicated due to using different measures of liquidity and whether liquidity is treated as a characteristic or risk factor. Pastor and Stambaugh (2003) find that expected returns are related to liquidity risk, the sensitivities of returns to fluctuations in aggregate liquidity, and they measure liquidity as the reversal of the price impact of trading volume. When examining relation between liquidity and return, Amihud (2002) and Acharya and Pedersen (2005) measure liquidity as the ratio of average absolute returns to trading volume, i.e., the Amihud illiquidity measure, Hasbrouck (2009) uses the effective spread measures, and Sadka (2006, 2010) investigates the role of liquidity risk in stock returns and measures liquidity based on price impact factors. Another issue is that the primary testing in this line of literature is the standard asset pricing tests that ignore causality and endogeneity.

Another line of literature relates liquidity to firm performance. Liquid markets permit non-blockholders to intervene and become blockholders, which then improves monitoring and corporate governance (Maug, 1998). Edmans et al. (2013) find that stock liquidity facilitates blockholder governance through the mechanisms of exit or voice. Using Real Estate Investment Trusts (REITs), Cheung, Chung and Fung (2015) show that liquidity has a causal and positive effect on firm value, due to better governance through the channel of institutional ownership. Liquidity can also have a disciplinary effect on management by facilitating informed selling or "dumping" ("Wall Street

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Walk") (e.g., Edmans, 2009; Admati and Pfleiderer, 2009). Subrahmanyam and Titman (2001) and Khanna and Sonti (2004) show that liquidity can positively affect firm performance by stimulating the entry of informed investors who make price more informative to stakeholders. Fang, Noe, and Tice (2009) show that stock market liquidity positively impacts firm value through increasing the information content of market prices and of performance-sensitive managerial compensation contracts.

Besides liquidity, algorithmic trading affects other aspects of the stock market, such as volatility. The empirical evidence of the effect of algorithmic trading on volatility is limited and mixed. For example, Brogaard (2011) finds that high frequency trading reduces intraday volatility in the short-term, while Zhang (2010), Scholtus, van Dijk, and Frijns (2014), and Boehmer, Fong, and Wu (2018) find that algorithmic trading increases volatility. Whether algorithmic trading positively or negatively affects volatility, it can impact asset prices through affecting volatility. Merton (1987) predicts that idiosyncratic risk is positively related to the expected stock returns when market is information-segmented and investors are under-diversified. Fu (2009) empirically confirms the positive relation between idiosyncratic volatility and expected return.[5] Lower idiosyncratic volatility would lead to lower risk and lower expected return or cost of capital, resulting in higher contemporaneous stock prices and firm value. If algorithmic trading reduces idiosyncratic volatility, greater algorithmic trading should increase firm value.

Another dimension through which algorithmic trading may affect firm value is its impact on return distributions, specifically return skewness. The asset pricing literature suggests that skewness is priced. For example, Mitton and Vorkink (2007) introduce a rational model of heterogeneous

[5] Other studies, e.g., Ang, Hodrick, Xing, and Zhang (2006), find negative relation between idiosyncratic volatility and expected return, the idiosyncratic volatility puzzle. Fu (2009) suggests that the negative relation between idiosyncratic volatility and expected return found in Ang, Hodrick, Xing, and Zhang (2006) is largely due to return reversal of a subset of small stocks with high idiosyncratic volatilities.

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preference for skewness and show that idiosyncratic skewness is priced. Some investors ("lotto investors") prefer lottery stocks, i.e., stocks with positive skewness and place a higher value on such stocks. So, greater idiosyncratic skewness is related to lower expected return and higher current stock price (and firm value), which has been confirmed empirically (see, e.g., Boyer, Mitton and Vorkink, 2010; Conrad, Dittmar and Ghysels, 2013). If algorithmic trading increases idiosyncratic skewness, greater algorithmic trading should increase firm value.

Thus, we test the following main hypotheses:

H1: All else equal, stocks that are more heavily traded using computer algorithms have higher liquidity.

H2: All else equal, stocks that are more heavily traded using computer algorithms have lower idiosyncratic volatility.

H3: All else equal, stocks that are more heavily traded using computer algorithms have greater idiosyncratic skewness.

H4: All else equal, firms whose stocks are more heavily traded using computer algorithms have higher value.

3. Data

3.1. Sample Selection

We start with a sample of all stocks covered in the TAQ database from 2002 to 2013. We limit our attention to the post-decimalization period because decimalization in 2001 was a major structural change to the stock trading system. TAQ contains intraday transaction (quotes and trades) data for all securities listed on NYSE, American Stock Exchange (AMEX), Nasdaq, and SmallCap issues. We obtain firm financial data from the Compustat database, stock return and total trading

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volume data from the Center for Research in Security Prices (CRSP), and analyst coverage data from the Institutional Brokers' Estimate System (I/B/E/S) database. We include only common stocks (those with a share code of 10 or 11 in the CRSP database). Restricted by firm-level variables, our sample is at quarter frequency. We also require each firm-fiscal quarter observation to have a non-missing value of Tobin's q, the measure of firm value, which we define below. Our final sample has 114,072 firm-quarter observations with 4,837 unique firms.

3.2. Variable Construction

3.2.1. Algorithmic Trading

Following Hendershott, Jones, and Menkveld (2011) and Boehmer, Fong, and Wu (2018), we construct a proxy measure for algorithmic trading based on electronic message traffic. Hendershott et al. (2011) have access to order-level messages, and their algorithmic trading proxy is the number of messages normalized by dollar volume. Specifically, they define the algorithmic trading proxy as the negative of dollar volume divided by the number of messages, including order submissions, order cancellations and trades. They also use the number of messages normalized by number of trades or the raw message as proxies and find the same results. Boehmer et al. (2018) construct a similar proxy. Because they do not have access to order-level messages and only observe each exchange's best quotes and trades, they define a message as either a trade or an update in the best bids and offers (BBO) and measure algorithmic trading as the negative of dollar volume associated with each message.[6]

The TAQ database that we use in this study contains transaction-level messages, but no order-level messages. Thus, we follow Boehmer et al. (2018) to define a message as either a trade

[6] Boehmer et al. (2018) point out that they can qualitatively replicate Hendershott et al. (2011) results using the U.S. portion of their sample and their algorithmic trading proxy constructed from the TAQ data.

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or a change in BBO, and we construct our algorithmic trading proxy measure as the normalized number of messages. To avoid a mechanical correlation between our algorithmic trading proxy and Tobin's q, we use number of trades instead of dollar volume to normalize the number of messages. Specifically, for firm *i*'s stock, its BBO is determined following Hasbrouck (2010),[7] and the number of trades and number of changes in BBO are aggregated over firm *i*'s fiscal quarter. Our algorithmic trading proxy, *Algo*, is defined as the negative of number of trades divided by the number of messages that includes changes in BBO and trades. Greater *Algo* indicates more messages per trade and higher algorithmic trading intensity.

Moreover, the TAQ database allows us to observe all quotes, not just BBO. Therefore, as a robustness check for *Algo*, we construct an alternative algorithmic trading proxy, *Algo_q*, by considering updates of all quotes, rather than only the best quotes, as messages. We define *Algo_q*, as the negative of number of trades divided by the number of messages that includes updates in quotes and trades.[8] These two measures have a correlation of 0.55, which suggests that they have a strong common component, yet have some independent variation.

3.2.2. Other Variables

We use Tobin's q to measure firm value. Following Roll, Schwartz, and Subrahmanyam (2009), we compute Tobin's q as the ratio of the market value of assets to the book value of assets of the firm, where the market value of assets is calculated as the sum of the market value of common stock, the liquidation value of preferred shares, and the book value of long-term debt.

[7] Following Hasbrouck (2010), we use all quotes available at a given time to determine the BBO, and do not drop quotes. If there are multiple quotes occurring in the same second, BBO is determined based on the order of quote records, and there can be updates in BBO in the same second.

[8] There are some stocks that are thinly traded but have relatively large numbers of quote updates, and therefore, have high values of *Algo* or *Algo_q*. Excluding those stocks by adding a minimum trade filter (e.g., 1,000 or 3,000 trades per year) for stocks to enter the sample does not change the results qualitatively.

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Stock liquidity is measured by effective spread (*Spread*), defined as the difference between the execution price and the mid-point of the prevailing bid-ask quote (Lee, Mucklow, and Ready, 1993). For each stock, we first calculate a volume-weighted daily average of effective spread and then average it across all trading days in a quarter. We use idiosyncratic volatility (*IVol*) to measure stock return volatility,[9] with *IVol* defined as the standard deviation of the residuals from fitting the four-factor model (Fama-French (1993) three factors plus the momentum factor) to daily returns over a quarter.[10] Idiosyncratic skewness (*ISkew*) is computed following Boyer, Mitton and Vorkink (2010). For stock *i* in the quarter *t*, its idiosyncratic skewness, $ISkew_{i,t,}$ is defined as

$$ISkew_{i,t} = \frac{\sqrt{n(n-1)}}{n-2}\left[\frac{\frac{1}{n}\sum_{d=1}^{n}\varepsilon_{i,d}^{3}}{\left(\frac{1}{n}\sum_{d=1}^{n}\varepsilon_{i,d}^{2}\right)^{3/2}}\right]$$

where, *n* is the number of trading days in the quarter *t*, $\varepsilon_{i,d}$ is the daily residual return of stock *i* on day *d* in the quarter *t*, with the daily residual return estimated from fitting the four-factor model to daily returns over the quarter.

We include a wide range of control variables in our analysis. We control for stock return and turnover to capture factors from the stock market that may affect firm value. Stock return (*Return*) is the accumulated daily return over a firm's previous fiscal quarter. Turnover (*Turnover*) is the total trading volume over a firm's fiscal quarter divided by the number of shares outstanding at the beginning of the quarter. In alternative specifications, turnover is replaced by total trading volume (*Volume*) or dollar volume (*Dollar Volume*).

[9] The results are qualitatively the same when we use volatility of raw daily returns over the fiscal year instead of idiosyncratic volatility.

[10] Our measure of IVol is computed similarly to Ang et al. (2006). Fu (2009) shows that the negative relation between IVol and expected return found by Ang et al. (2006) is driven mainly by return reversals of a subset of small stocks with high IVol, and that the true relation should be positive. To ensure that our measure of IVol is positively related to future returns, we perform a test using our sample. We form decile portfolios each quarter based on lagged IVol, and compute return of each portfolio. Then we compute the time series average of portfolio returns. We find a positive relation between lagged IVol and return. The portfolio return increases when moving from the lowest IVol portfolio to the highest IVol portfolio, with the return of the highest IVol portfolio more than double that of the lowest IVol portfolio.

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Following Roll, Schwartz, and Subrahmanyam (2009) and Fang, Noe, and Tice (2009), we also control for several firm characteristics that are related to firm valuation, measured over the current fiscal quarter. Return on assets (*ROA*) is net income divided by the book value of assets. *Size* is measured as the natural logarithm of total book assets. *Leverage* is total debt divided by book value of assets. *Analyst* is the natural logarithm of one plus the number of analysts following a firm. *Dividend* is an indicator variable for whether the firm pays a dividend during the fiscal quarter. *SP500* is an indicator variable for whether the firm is included in the S&P 500 during the fiscal quarter. Industries are defined based on the Fama and French 48 industry classifications.

3.2.3. Summary Statistics

Table 1 Panel A presents summary statistics for the sample. In our sample, the mean value of *Tobin's q* is 1.955, with a standard deviation of 1.829 and median of 1.430. For the average stock over our sample period, the average turnover is 1.544, with average trading volume of 39.20 million shares and dollar volume of $3.459 billion. The mean value of the algorithmic trading measure, *Algo*, is -0.348, which means that for the average stock, there are 0.348 trades per electronic message, or equivalently, 2.87 messages per trade.

Table 1 Panel B reports summary statistics of the algorithmic trading measure, *Algo*, year by year over the whole sample period. With the implementation of the Regulation National Market System during 2007, which incentivizes electronic execution across markets, algorithmic trading intensity increases dramatically from 2006 to 2008. Interestingly, the time series is relatively flat before 2007 and after 2007. The *t*-test result reported at the lower portion of Panel B shows that algorithmic trading intensity is significantly higher after 2007.

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4. Results

4.1. Baseline Results

The (baseline) results in this section ignore potential endogeneity concerns; in a later section, we use an exogenous shock to algorithmic trading to address such concerns. We hypothesize that algorithmic trading impacts various aspects of stock trading and as a result, increases firm value. Thus, we start our analysis by examining the relation between algorithmic trading and stock liquidity (measured by effective spread), idiosyncratic volatility, and idiosyncratic skewness. Specifically, for firm *i* and year-quarter *t*, we run the following panel regression model,

$$
\begin{aligned}
Spread_{i,t}\ (IVol_{i,t} \text{ or } ISkew_{i,t}) = {} & a + b\times Algo_{i,t} \\
& + c\times Turnover_{i,t} \text{ (alternately, } Volume_{i,t} \text{ or } Dollar\ Volume_{i,t}) \\
& + d\times ROA_{i,t} + e\times Size_{i,t} + f\times Analyst_{i,t} + g\times Return_{i,t-1} \\
& + firm\ FE + industry\ FE + year{*}quarter\ FE + \text{error term} \qquad (1)
\end{aligned}
$$

where *Spread* is effective spread, *IVol* is idiosyncratic volatility, *ISkew* is idiosyncratic skewness, *Algo* is the proxy for algorithmic trading intensity, *Turnover*, *Volume* (total trading volume), or *Dollar Volume* (total dollar volume) are controlled in separate specifications, respectively, and the control variables are defined in the previous section. Firm fixed effects (firm dummies), industry dummies and year-quarter dummies are included in the regressions.

The panel regression results are shown in Table 2. When the dependent variable is effective spread, in each specification, the coefficient on *Algo* is negative and statistically significant at the 1% level. The significantly negative relation between algorithmic trading and effective spread suggests that stocks subject to higher algorithmic trading intensity have narrower effective spreads, i.e., higher liquidity, consistent with the Hypothesis 1. The negative (positive) relation between

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algorithmic trading and effective spread (stock liquidity) is also economically significant. The coefficient on *Algo* suggests that a one standard deviation increase in *Algo* is associated with a reduction in effective spread by about 5% of its sample mean.[11]

When the dependent variable is idiosyncratic volatility, the coefficient on *Algo* is negative and statistically significant at 1% level in all specifications. The significantly negative relation between algorithmic trading and idiosyncratic volatility suggests that stocks subject to higher algorithmic trading intensity have lower idiosyncratic volatility, consistent with the Hypothesis 2. With idiosyncratic skewness as the dependent variable, the coefficient on *Algo* is significantly positive in all specifications, suggesting the stocks subject to higher algorithmic trading intensity have larger idiosyncratic skewness, consistent with the Hypothesis 3. The negative (positive) relation between algorithmic trading and idiosyncratic volatility (skewness) is economically significant. Using the results from the specification (1) as examples, the coefficients on *Algo* suggests that a one standard deviation increase in *Algo* is associated with a reduction in *IVol* by about 4% of its sample mean or an increase in *ISkew* by a magnitude about equal to its sample median.

After having established the significant relation between algorithmic trading and stock liquidity, idiosyncratic volatility, and idiosyncratic skewness, we next examine the relation between algorithmic trading and firm value, measured by Tobin' q.[12] For firm *i* and year-quarter *t*, the following panel regressions are run:

[11] The sample standard deviation of *Algo* is 0.276 and the sample mean of effective spread is 0.0056.

[12] The relation between algorithmic trading and firm value can be contemporaneous if it works through the asset pricing channel or a long-term process through the corporate finance channel. In the main analysis Tobin's q and algorithmic trading are measured in the same year-quarter, while in a robustness check, we show that the main results hold when the algorithmic trading measure is lagged by one quarter or one year.

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$$Tobin's\ q_{i,t} = a + b\times Algo_{i,t} + c\times Turnover_{i,t} \text{ (alternately, } Volume_{i,t} \text{ or } Dollar\ Volume_{i,t}) + d\times ROA_{i,t} + e\times Size_{i,t} + f\times Leverage_{i,t} + g\times Analyst_{i,t} + h\times Return_{i,t} + i\times Dividend_{i,t} + j\times SP500_{i,t} + firm\ FE + industry\ FE + year*quarter\ FE + \text{error term} \quad (2)$$

Equation (2) is estimated with firm fixed effects controlled. Although Equation (2) controls for a series of variables related to stock market and firm characteristics, it is possible that some unobservable firm characteristics may affect algorithmic trading activity and also correlate with firm value, and such omitted variables can bias estimation results of a simple ordinary least square regression. Thus, we use firm fixed effects to control for the unobservable firm characteristics that are constant over time. We also control for year-quarter and industry fixed effects to capture factors such as macroeconomic conditions and industry-specific factors that may affect all firms. All standard errors are adjusted for clustering at the firm level.

The results of the panel regressions in Equation (2) are shown in Table 3, models 1 to 3. For the three specifications that control for turnover, total trading volume, and total dollar volume, respectively, the coefficients on *Algo* are all positive and significant at the 1% level, implying that firms whose stocks have higher algorithmic trading intensity have greater firm value, consistent with Hypothesis 4. The result is also economically significant. For example, based on the standard deviation of *Algo* and the sample mean of Tobin's q, the coefficient on *Algo* in model 3 implies that a one standard deviation increase in *Algo* results in an increase in Tobin's q by 0.108, which is about 5.5% of its sample mean.

Most of the control variables are significantly related to Tobin's q. The coefficients on volume (in model 2) and dollar volume (in model 3) are significantly positive, although the coefficient on turnover (in model 1) is negative. As expected, stock return is positively related to firm value. Firm size has a significantly negative coefficient, implying that on average smaller firms

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have higher Tobin's q, which is consistent with the findings in Fang, Noe, and Tice (2009).[13] Contrary to expectation, return on asset has a significantly negative effect. Analyst coverage has a significantly positive coefficient suggesting that analyst coverage may create attention and lead to higher firm valuation. Both the dividend dummy and the SP500 dummy have significantly positive coefficients. Dividend-paying firms may have higher valuation than non-dividend paying firms potentially because they are less financially constrained (Fazzari, Hubbard, and Petersen, 1988) and have lower agency costs of free cash flow or overinvestment (Rozeff, 1982; Easterbrook, 1984; Jensen, 1986). Leverage has a positive effect, possibly due to the disciplining effect of debt and the associated lower agency cost in firms with higher leverage. S&P 500 firms may have higher value than non-S&P 500 firms because they have greater liquidity.

In summary, the results in this section indicate that algorithmic trading activity has a significant relation with stock liquidity, idiosyncratic volatility, and idiosyncratic skewness, all in directions that are consistent with a positive impact on share price. We also document a positive relation directly between algorithmic trading activity and firm value after controlling for other variables related to firm value, as well as firm, industry and year fixed effects. The results are consistent with Hypotheses 1 through 4 and imply that algorithmic trading increases firm value through impacting various stock characteristics.

4.2. Causal Evidence and the Underlying Channels

The results in the previous section demonstrate a strong relation between algorithmic trading intensity and various stock characteristics as well as firm value. Stocks subject to higher algorithmic trading intensity have higher liquidity, lower idiosyncratic volatility and greater idiosyncratic

[13] In this paper and in Fang, Noe, and Tice (2009), firm size is measured by natural logarithm of book value of assets, which is negatively related to Tobin's q. Roll, Schwartz, and Subrahmanyam (2009) measure firm size by market capitalization, which is positively related to Tobin's q, and find a positive relation between Tobin's q and firm size.

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skewness, and also have higher firm value. Algorithmic trading may increase firm value through its impacts on liquidity, volatility, and skewness, but the causality may also run in the opposite direction if, for example, more liquid or positively skewed stocks and firms with higher value attract more algorithmic trading. In this section we employ an exogenous shock to algorithmic trading to address concerns of reverse causality and endogeneity and identify channels through which algorithmic trading affects firm value.

4.2.1. Causal Evidence from the Auto Quote

We follow Hendershott, Jones, and Menkveld (2011) and Boehmer, Fong, and Wu (2018) in using the change to automated quote dissemination (auto quote) by the New York Stock Exchange (NYSE) in 2003 as an exogenous shock to algorithmic trading intensity. Hendershott, Jones, and Menkveld (2011) show that auto quote is positively correlated with their electronic message-based algorithmic trading measure, and they use auto quote as an instrument for algorithmic trading. Auto quote occurred to NYSE stocks from January 2003 to May 2003. Auto quote was an important innovation for algorithmic traders. Specialists used to manually disseminate the inside quote, whereas with auto quote, a new automated quote is generated immediately whenever there is a change to the NYSE limit order book. An automated quote update could provide more immediate feedback about the potential terms of trade and more critical new information to algorithms, which could result in more electronic message traffic and higher algorithmic trading intensity. Thus, this exogenous shock in market structure provides quicker feedback to algorithms, potentially increasing the amount of algorithmic trading and resulting in more electronic message traffic.[14,15]

[14] See Hendershott, Jones, and Menkveld (2011) for more discussion about the NYSE automated quote dissemination.
[15] The mean values of *Algo* for NYSE stocks are -0.470 and -0.402 in 2002 and 2004, respectively, corresponding to 2.12 and 2.49 messages per trade in 2002 and 2004, respectively..

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To employ the advent of auto quote as an exogenous shock to algorithmic trading, we perform a two-stage least squares regression analysis over a narrow, two-quarter window surrounding the advent of auto quote, i.e., the quarter before Jan. 2003 and the quarter after May 2003. Moreover, since the auto quote only affected NYSE stocks, the tests are limited to NYSE stocks. We attempt to establish a causal effect of algorithmic trading on firm value by using the time-series variation in algorithmic trading that is attributed to the exogenous shock of auto quote.

Specifically, in the first stage, *Algo* is regressed on an auto quote dummy variable, *Post*, and stock and firm characteristic variables. *Post* is set to one for the quarter after the auto quote (i.e., the first quarter after May 2003) and zero for the quarter before the auto quote (i.e, the last quarter of 2002). The fitted value of *Algo*, *Fitted Algo*, computed from the first stage regression is then included as the main independent variable of interest in the second stage regression of Tobin's q, where other independent variables include the control variables in equation (2) and effective spread, idiosyncratic volatility and idiosyncratic skewness. To the extent that the fitted value of *Algo* captures the exogenous variation in algorithmic trading due to the auto quote, the result from the second stage regression provides evidence of a causal effect of algorithmic trading on firm value.

Table 4 reports the results of the first stage regressions in which the dependent variable is *Algo*. The coefficient on the auto quote dummy, *Post*, is highly significant and positive, confirming that auto quote has a significantly positive impact on algorithmic trading. The first-stage F-statistics are all above 10 in each model, indicating strong instruments. The fitted value of *Algo*, *Fitted Algo*, is calculated from the first stage regressions and then used in the second stage regressions.

The results of the second stage regressions are shown in Table 5. The dependent variable is Tobin's q. There are three regression models in each panel, corresponding to the three measures of overall stock trading intensity. In Panel A, the regression models are the same as those in Table 3,

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except for the algorithmic trading proxy, *Algo*, being replaced by the fitted value of *Algo*, *Fitted Algo*. The coefficients on *Fitted Algo* are all positive and significant at 1% level across the three models. Since *Fitted Algo* captures the exogenous variation in algorithmic trading attributed to the auto quote, the results shown in Panel A of Table 5 suggest a causal (and positive) effect of algorithmic trading on firm value.

Since algorithmic trading has a significantly positive effect on firm value and the auto quote induces an exogenous increase in the intensity of algorithmic trading of the NYSE stocks, we expect positive price reactions by those stocks. To confirm the conjecture, we compute the cumulative abnormal return, *CAR*, around the auto quote phase-in dates of the stocks.[16] The abnormal daily returns are calculated as the difference between actual returns and expected returns estimated using the market model with the market proxied by the value-weighted CRSP index.[17] The expected returns are estimated using daily returns over the benchmark period between 250 days and 20 days prior to the phase-in date. We compute *CAR* over several different event windows around the phase-in date *t*, and find significantly positive *CAR* in all cases. For example, the mean values of *CAR*[*t*-3, *t*+3], *CAR*[*t*-5, *t*+5], *CAR*[*t*-5, *t*+10] and *CAR*[*t*-5, *t*+30] are 1.4%, 1.8%, 2.6% and 4.2%, respectively, and are significantly positive with *t*-test statistics of 7.35, 6.86, 7.63 and 8.68, respectively. The magnitude of *CAR* around the NYSE auto quote is comparable to the reported stock market reactions to improvements in market microstructure. For example, Amihud, Mendelson and Lauterbach (1997) find about 5.5% *CAR*[*t*-5, *t*+30] around the introduction of a new trading mechanism in the Tel Aviv Stock Exchange, which transferred stocks from trading once a day in a call auction to continuous trading sessions following an opening daily call auction.

[16] The detailed auto quote phase-in dates are from Terrence Hendershott's website. We thank Terrence Hendershott for making the data available.

[17] The results are qualitatively the same when the Fama-French three-factor model is used to compute the expected returns.

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4.2.2. Underlying Channels

We next provide further evidence on the underlying channels through which algorithmic trading impacts firm value. In the three regression models in Panel B of Table 5, we replace *Fitted Algo* by effective spread (*Spread*), idiosyncratic volatility (*IVol*) and idiosyncratic skewness *(ISkew*) to investigate the impact of these stock characteristics on firm value. The coefficients on *Spread* are significantly negative at 1% level in all three models, indicating a positive effect of stock liquidity (negative effect of effective spread) on firm value. The coefficients on *IVol* are also significantly negative at 1% level in all three models, indicating a negative effect of idiosyncratic volatility on firm value. The coefficients on *ISkew* are significantly positive, indicating a positive effect of idiosyncratic skewness on firm value. The results are consistent with prior literature discussed earlier in the paper (e.g., Fang, Noe, and Tice, 2009; Fu, 2009; Boyer, Mitton and Vorkink, 2010; Conrad, Dittmar and Ghysels, 2013).

We then include *Fitted Algo*, *Spread*, *IVol* and *ISkew* in the regression models in Panel C. The coefficients on *Spread*, *IVol* and *ISkew* remain significant and have the same signs as those in Panel B. The coefficients on *Fitted Algo* are still significantly positive as those in Panel A. However, compared to those in Panel A, the magnitudes of the coefficients on *Fitted Algo* are much smaller in Panel C when *Spread*, *IVol* and *ISkew* are added to the regressions. Adding the three variables to the Tobin's q regressions reduces the effect of *Fitted Algo* on firm value by about two-thirds, which implies that about two-thirds of the positive impact of algorithmic trading on firm value is due to its impact on effective spread~~,~~ (i.e., stock liquidity), idiosyncratic volatility, and idiosyncratic skewness. The results are consistent with the hypothesis that algorithmic trading positively affects firm value through the channels of stock liquidity, idiosyncratic volatility, and idiosyncratic skewness.

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In addition, to provide further support for the underlying channels of stock liquidity, idiosyncratic volatility, and idiosyncratic skewness, we test the causal effect of algorithmic trading on these three stock characteristics. We run the two stage regression analysis surrounding the advent of auto quote of NYSE stocks. The first-stage regressions remain the same as those in Table 4, and the second-stage regressions are similar to those in Table 2, with *Algo* replaced by the *Fitted Algo* from the first-stage regressions. For brevity, the results from the second-stage regressions are not tabulated. *Fitted Algo* is significantly and negatively related to effective spread and idiosyncratic volatility and is significantly and positively related to idiosyncratic skewness. Thus, algorithmic trading significantly increases stock liquidity and idiosyncratic skewness and decreases idiosyncratic volatility, which translate to higher firm value.

To assess the relative importance of each of the stock liquidity, idiosyncratic volatility, and idiosyncratic skewness channels, we perform the second-stage firm value regression analyses as shown in Table 6. The dependent variable is Tobin's q in all models. *Fitted Algo* is the main independent variable of interest in the model (1), with *Spread*, *IVol* and *ISkew* included in the models (2), (3) and (4), respectively. The model (5) includes all of three variables. Turnover is controlled in Table 6, and the results are similar when Volume or Dollar Volume are controlled. By comparing each of the models (2), (3) and (4) with the model (1), we estimate what fractions of the algorithmic trading's effect on firm value are from the channels of stock liquidity, idiosyncratic volatility, and idiosyncratic skewness. The results from the models (1) and (2) indicate that adding *Spread* to the firm value regression reduces the coefficient on *Fitted Algo* by about 50%.[18] Similarly, in comparison to those in the model (1), the results from the models (3) and (4) suggest that adding *IVol* or *ISkew* to the firm value regression reduces the coefficient on *Fitted Algo* by

[18] The result is largely unchanged when the model (2) of Table 6 additionally controls for other aspects of liquidity such as price impact, measured by Amihud illiquidity (Amihud, 2002) or the volatility over volume liquidity proxies (Fong, Holden, and Tobek, 2018).

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about 28% and 6%, respectively. *Spread*, *IVol* and *ISkew* are all included in the model (5), where the coefficient on *Fitted Algo* falls by about 68%. The stock liquidity channel is the main mechanism by which algorithmic trading affects firm value, followed by the idiosyncratic volatility channel and the idiosyncratic skewness channel. The three channels altogether account for about 68% of the positive effect of algorithmic trading on firm value.

Overall results in this section provide causal evidence that algorithmic trading positively impacts firm value and suggest that algorithmic trading affects firm value through impacting stock liquidity, idiosyncratic volatility and idiosyncratic skewness.

4.3. Nonlinearity of Algorithmic Trading's Effects

Considering the mixed evidence in the literature on algorithmic trading's impact on the market quality, we conjecture that algorithmic trading may have nonlinear effects on firm value. So, we next turn to investigate how the effects of algorithmic trading on firm value depend on firm characteristics and time periods.

Hendershott, Jones, and Menkveld (2011) find that the positive impact of algorithmic trading on stock liquidity is stronger for large and liquid stocks that have high intensity of algorithmic trading. Boehmer, Fong, and Wu (2018) find similar results in the international setting. Motivated by those findings, we expect that the positive effects of algorithmic trading on liquidity and firm value are stronger when algorithmic trading intensity is higher, such as in cases of firms with more liquid stocks and larger firms. In addition, as shown in Table 1 Panel B, algorithmic trading intensity increased substantially during 2007 due to the implementation of Reg NMS between 5/07 and 10/07 and is significantly higher after 2007 than before 2007. Thus the effect of algorithmic trading is expected to be stronger after 2007. Therefore, we examine the effect of

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algorithmic trading on firm value across groups of firms based on stock liquidity and firm size, and across different time periods, i.e., before and after 2007.

For that purpose, we define three dummy variables. Each quarter, firms in each industry are divided into two groups based on stock liquidity (effective spread) and firm size. *Hispread* is a dummy variable that is equal to one for firms with effective spread higher than the industry mean in the quarter, and zero otherwise. *Large* is a dummy variable that is equal to one for firms with size greater than the industry mean in the quarter, and zero otherwise. The sample period is divided into two sub periods, with the dummy variable *After07* equal to one for the sub period after 2007 (2008 to 2013) and zero for the sub period before 2007 (2002 to 2006). Each dummy variable and its interaction with *Algo* are then added to Equation (2), with the estimation results reported in Table 7.[19] The coefficient on *Algo×Hispread* is significantly negative while the coefficients on *Algo×Large* and *Algo×After07* are significantly positive, indicating that algorithmic trading has a stronger positive impact on firm value for firms with more liquid stocks, larger firms, and in the post-2007 period. Overall, the results are consistent with the notion that algorithmic trading has a stronger positive impact on firm value when algorithmic trading intensity is higher.[20]

We also confirm the nonlinear effect of algorithmic trading on firm value by adopting the two-stage least squares regression approach and using the subsample of NYSE stocks around auto quote implementation. The first stage regressions are the same as those in Table 4, where *Algo* is regressed on the auto quote dummy, *Post*, and controls variables. The fitted value of *Algo*, *Fitted Algo*, is then used to examine how the algorithmic trading's effect on firm value depends on stock liquidity and firm size; the dependence of the algorithmic trading's effect on time periods cannot be

[19] Table 7 shows results of regressions controlling for turnover (*Turnover*) as the measure of overall stock trading. The results on the interaction terms are very similar when stock trading volume and dollar volume are controlled.

[20] We also examine the dependence of algorithmic trading's effect on liquidity and find similar results. Untabulated results show that algorithmic trading has a stronger positive (negative) impact on stock liquidity (effective spread) for firms with more liquid stocks, larger firms, and in the post-2007 period.

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examined using the subsample around auto quote. The results are reported in Table 8. The regressions are similar to the panel regression model (2), with *Fitted Algo* replacing *Algo* and its interactions with *Hispread* or *Large* included. The coefficients on *Fitted Algo* are significantly positive, while the coefficients on *Fitted Algo* ×*Hispread* are significantly negative and the coefficients on *Fitted Algo*×*Large* are significantly positive. The results suggest that the positive impact of algorithmic trading on firm value is stronger for firms with more liquid (smaller spread) stocks and for large firms.

In short, the overall results reported in this section indicate that the positive impacts of algorithmic trading on firm value tend to be stronger when algorithmic trading intensity is higher.

4.4. Robustness Checks

The results in previous sections show that the finding of a positive effect of algorithmic trading on firm value is robust to the inclusion of year-quarter, firm and industry fixed effects, and the control for reverse causality. This section performs additional robustness checks.

4.4.1. Lagged Algorithmic Trading Measure

The results reported above are based on the data with quarterly frequency. One concern would be that it may take some time for algorithmic trading to impact firm value. For example, the liquidity channel may work through the effect of stock liquidity on managerial incentives (Fang, Noe, and Tice, 2009), which is a long-term process. To address that concern, we repeat the regression analyses in Table 3 with the algorithmic trading measure, *Algo*, being lagged one quarter. The results are reported in Table 9, Panel A, and show a significantly positive effect of lagged *Algo*

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on Tobin's q. We also constructed a sample on annual frequency, and the untabulated results show that our findings hold in the annual frequency data and with *Algo* being lagged one year.

4.4.2. Alternative Proxies for Algorithmic Trading

Recognizing that there will be measurement error in our measure of algorithmic trading, we used alternative definitions of algorithmic trading to determine if our results were sensitive to our definition. Rosu, Sojli and Tham (2018) study the relation between quoting activity, measured by the quote-to-trade (QT) ratio, and the cost of capital. Their QT ratio measure is close in spirit to the algorithmic trading measure in Hendershott et al. (2011). Thus, we repeat the regression analyses in Table 3 using the QT ratio based proxy for algorithmic trading, *Algo_QT*, which is the ratio of the number of quote updates at the national BBO to the number of trades. The results are reported in Table 9, Panel B, and confirm that our main results are robust to the use of the QT ratio measure.

In the main analyses, we follow Boehmer et al. (2018) to define an electronic message as either a trade or a change in BBO, and construct the proxy *Algo* as the negative of number of trade normalized messages. However, algorithmic traders may submit orders both inside and outside the quotes. So, alternatively, we consider updates of all quotes, not just the best quotes, as messages, and define a message as either a trade or a change in quotes. We then define the alternative proxy for algorithmic trading, *Algo_q* as the negative of number of trades divided by the number of messages that include the number of updates in quotes and the number of trades. We repeat the regression analyses in Table 3 using *Algo_q* to measure algorithmic trading activity. The untabulated results show that our main results are robust to the use of the alternative proxy for algorithmic trading.

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4.4.3. Change in Algorithmic Trading

We also estimate first difference regressions. Specifically, we compute the first differences for all variables in Table 3 and regress the change in Tobin's q on the change in *Algo* (C*h_Algo*) and the change in control variables. The results are shown in Table 10. The coefficients on *Ch_Algo* are significantly positive, consistent with positive impact of algorithmic trading on firm value.

4.4.4. Rank Regressions

Another issue to be addressed is the non-normality and skewness of the distributions of several dependent and independent variables. Summary statistics reported in Table 1 show that *Tobin's q*, *Turnover*, *Volume*, and *Dollar Volume* are right skewed. To address this concern, we use a rank regression approach. All variables in Equation (2), except for the dummy variables, are ranked from low to high in the 0 to 100 scale, and the ranks, instead of the original values, of the variables are used in regression analysis. The model specification is the same as Equation (2). Untabulated results show that the main results are robust to the use of rank regressions.

5. Conclusion

While algorithmic trading is criticized for creating excess volatility and destabilizing stock markets, some evidence suggests that algorithmic trading positively impacts market quality by increasing liquidity, reducing idiosyncratic volatility, and contributing to price discovery. We contribute evidence to the debate by examining the effect of algorithmic trading activity on firm value.

Using an algorithmic trading proxy based on electronic message traffic, we find significant effects of algorithmic trading on three stock characteristics and firm value. By exploiting the NYSE

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implementation of auto quotation in 2003 as an exogenous shock to algorithmic trading and using a two-stage least squares regression framework, we provide evidence of a causal effect of algorithmic trading on firm value and show that algorithmic trading positively impacts firm value through improving stock liquidity, reducing idiosyncratic volatility, and increasing idiosyncratic skewness. The result is robust to controlling for other variables related to firm value, the use of various specifications, inclusion of year-quarter, industry and firm fixed effects, and the use of an alternative proxy for algorithmic trading. We document that 68% of the positive impact of algorithmic trading on firm value is through changes in liquidity, idiosyncratic volatility, and idiosyncratic skewness, suggesting there are still other dimensions through which algorithmic trading positively impacts firm value. In addition, we show that the cumulative abnormal returns attributed to the increase in algorithmic trading around the implementation of auto quote are similar in magnitude to those found by Amihud, Mendelson, and Lauterbach (1997) when stocks switched to a different trading structure at the Tel Aviv Stock Exchange. The positive effect of algorithmic trading on firm value is stronger for firms with higher stock liquidity, for larger firms, and in the post-2007 period when algorithmic trading intensity is higher.

This work is one of the first to attempt to directly measure the impact of the increase in algorithmic trading on firm value, and thus implicitly measure the impact on firm fundamentals. The overall results in this study suggest that algorithmic trading generates net benefits for firm value through impacting stock liquidity, idiosyncratic volatility and idiosyncratic skewnesss, and firms benefit more from algorithmic trading when algorithmic trading intensity is higher. Our results suggest there are opportunities to measure more directly the impact of algorithmic trading on other firm fundamentals that we know directly impact firm value.

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Table 1. Summary Statistics

Panel A. Summary Statistics of the Sample

The table reports summary statistics for the sample. Tobin's q is the market value of common stock plus liquidation value of preferred shares plus book value of long-term debt divided by the book value of assets. *Algo* is the proxy for algorithmic trading, defined over firm *i*'s fiscal quarter as the negative of number of trades divided by the number of message that includes the number of changes in best bid-offer (BBO) and the number of trades. Turnover is the trading volume of firm *i*'s stock over the fiscal quarter divided by the number of shares outstanding at the beginning of the fiscal quarter. Volume is the trading volume of firm *i*'s stock over the fiscal quarter (multiplied by 10^{-8}). Dollar Volume is the dollar trading volume of firm *i*'s stock over the fiscal quarter (multiplied by 10^{-9}). Effective Spread (Spread) is the difference between the execution price and the mid-point of the prevailing bid-ask quote. IVol is idiosyncratic volatility computed as the standard deviation of daily residual returns from fitting the four factor model. ISkew is idiosyncratic skewness. ROA is return on assets measured as net income divided by book value of assets. Firm size (Size) is measured by Ln(Total Assets). Leverage is measured as total debt divided by book value of assets. Analyst is analyst coverage measured by the natural logarithm of one plus the number of analysts following firm *i* during fiscal quarter. Stock Return (Return) is stock return over the previous fiscal quarter. Dividend is a dummy variable for whether the firm pays a dividend during the fiscal quarter. SP500 is a dummy variable for whether the firm is included in the S&P 500 during the fiscal quarter. The sample period is from 2002 through 2013. Only common stocks (those with a share code of 10 or 11 in the CRSP database) are included in the sample.

Variable	Mean	Std. Dev.	5th Pctl	25th Pctl	Median	75th Pctl	95th Pctl
(1) Tobin's q	1.955	1.829	0.613	1.002	1.430	2.264	5.025
(2) Algo	-0.348	0.276	-0.825	-0.556	-0.356	-0.062	-0.000
(3) Turnover	1.544	1.862	0.128	0.452	1.086	2.178	5.437
(4) Volume (10^8)	39.20	160	0.017	1.073	7.058	24.7	148
(5) Dollar Volume(10^9)	3.459	9.381	0.003	0.031	0.289	2.006	18.223
(6) Spread	0.006	0.011	0.000	0.001	0.002	0.006	0.05
(7) IVol	0.029	0.017	0.011	0.018	0.025	0.036	0.061
(8) ISkew	0.076	3.02	-5.269	-1.160	0.206	1.551	4.685
(9) ROA	0.036	0.219	-0.379	0.018	0.077	0.143	0.255
(10) Size	6.390	1.804	3.628	4.473	5.947	7.326	9.665
(11) Leverage	0.197	0.205	0	0.018	0.147	0.302	0.601
(12) Analyst	1.465	1.103	0	0	1.609	2.398	3.178
(13) Return	0.205	0.664	-0.611	-0.150	0.104	0.388	1.374
(14) Dividend	0.367	0.482	0	0	0	1	1
(15) SP500	0.107	0.309	0	0	0	0	1

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Table 1 Panel B. Year-by-year Summary Statistics of the Algorithmic Trading Measure

The table reports the summary statistics of the algorithmic trading measure, *Algo*, each year from 2002 to 2013 (upper portion) and the result of t-test of means of *algo* after 2007 vs. before 2007 (lower portion).

Year	Mean	Median	Std. Dev.
2002	-0.505	-0.497	0.142
2003	-0.521	-0.524	0.176
2004	-0.503	-0.518	0.202
2005	-0.531	-0.546	0.202
2006	-0.526	-0.535	0.227
2007	-0.375	-0.357	0.215
2008	-0.173	-0.081	0.217
2009	-0.193	-0.061	0.252
2010	-0.183	-0.038	0.246
2011	-0.166	-0.045	0.233
2012	-0.169	-0.053	0.223
2013	-0.192	-0.095	0.234
Before 2007 (2002 to 2007)	-0.517		
After 2007 (2008 to 2013)	-0.208		
Difference (After – Before)	0.309		
P value of t-test	0.000		

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Table 2. Effect of Algorithmic Trading on Stock Characteristics

The table reports panel regressions estimates of effect of algorithmic trading on stock characteristics. The dependent variable is effective spread (*Spread*), idiosyncratic volatility (*IVol*) or idiosyncratic skewness (*ISkew*). The independent variables include algorithmic trading proxy (*Algo*), Turnover (model (1)), Volume (model (2)), Dollar Volume (model (3)), ROA, firm size (*Size*), analyst coverage (*Analyst*), stock return (*Return*), firm dummies, industry dummies and year-quarter dummies. The industry classifications are defined by Fama and French (1997). Coefficients on firm, industry, and year-quarter dummies are not reported. T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is from 2002 through 2013. Only common stocks (those with a share code of 10 or 11 in the CRSP database) are included in the sample. See Table 1 for variable definitions.

	Spread			IVol			ISkew		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Algo	-0.001***	-0.001***	-0.001***	-0.004***	-0.003***	-0.003***	0.713***	0.653***	0.710***
	(-2.81)	(-3.03)	(-2.69)	(-6.55)	(-4.89)	(-6.22)	(12.44)	(11.11)	(12.36)
Turnover	-0.000***			-0.000			-0.000***		
	(-6.89)			(-1.39)			(-6.00)		
Volume		-0.000**			0.000***			-0.000***	
		(-2.09)			(5.46)			(-4.54)	
Dollar Volume			0.000			0.000***			-0.000
			(1.09)			(3.60)			(-0.94)
ROA	-0.005***	-0.005***	-0.005***	-0.013***	-0.013***	-0.013***	-0.210*	-0.215*	-0.210*
	(-8.05)	(-8.05)	(-8.04)	(-3.90)	(-3.87)	(-3.91)	(-1.77)	(-1.81)	(-1.76)
Size	-0.003***	-0.003***	-0.003***	-0.004***	-0.004***	-0.004***	0.039	0.047*	0.040
	(-17.06)	(-16.96)	(-17.10)	(-15.99)	(-16.44)	(-16.17)	(1.59)	(1.91)	(1.63)
Analyst	-0.001***	-0.001***	-0.001***	-0.000	-0.000	-0.000	-0.085***	-0.082***	-0.084***
	(-11.36)	(-11.32)	(-11.43)	(-0.58)	(-0.90)	(-0.70)	(-5.16)	(-4.98)	(-5.14)
Return	0.000	0.000	0.000	0.005***	0.005***	0.005***	5.035***	5.036***	5.035***
	(0.01)	(0.02)	(0.01)	(12.44)	(12.46)	(12.45)	(39.31)	(39.35)	(39.32)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	113,814	113,815	113,815	113,953	113,954	113,954	113,878	113,879	113,879
adj. R^2	0.251	0.251	0.251	0.358	0.361	0.359	0.225	0.225	0.225

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Table 3 Effect of Algorithmic Trading on Firm Value

The table reports panel regressions estimates of effect of algorithmic trading on firm value. The dependent variable is Tobin's q. The independent variables include algorithmic trading proxy (*Algo*), *Turnover* (model (1)), *Volume* (model (2)), *Dollar Volume* (model (3)), *ROA, Size, Leverage, Analyst, Return, Dividend, SP500,* firm dummies, industry dummies and year-quarter dummies. The industry classifications are defined by Fama and French (1997). Coefficients on firm, industry, and year-quarter dummies are not reported. T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is from 2002 through 2013. Only common stocks (those with a share code of 10 or 11 in the CRSP database) are included in the sample. See Table 1 for variable definitions.

	Tobin's q		
	(1)	(2)	(3)
Algo	0.323***	0.404***	0.390***
	(5.97)	(7.27)	(6.73)
Turnover	-0.000**		
	(-1.97)		
Volume		0.000***	
		(6.09)	
Dollar Volume			0.000***
			(3.24)
ROA	-0.707*	-0.699*	-0.708*
	(-1.74)	(-1.73)	(-1.75)
Size	-0.647***	-0.658***	-0.668***
	(-16.79)	(-17.06)	(-17.52)
Leverage	0.419*	0.424*	0.431**
	(1.92)	(1.94)	(1.98)
Analyst	0.266***	0.262***	0.260***
	(17.29)	(17.20)	(17.22)
Return	0.826***	0.825***	0.825***
	(24.84)	(24.80)	(24.86)
Dividend	0.092***	0.105***	0.099***
	(3.10)	(3.56)	(3.32)
SP500	0.389***	0.379***	0.376***
	(4.70)	(4.63)	(4.77)
Firm FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes
Observations	114,005	114,006	114,006
Adjusted R^2	0.162	0.165	0.172

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Table 4. First Stage Regression of Algorithmic Trading for NYSE Stocks around Auto Quote

This table reports the results of the first stage regressions of the two-stage regression analyses for NYSE stocks over the two quarters around the auto quote. The dependent variable is the algorithmic trading proxy, *Algo*. *Post* is a dummy variable that is set to one for the quarter after the auto quote (i.e., the first quarter after May 2003) and zero for the quarter before the auto quote (i.e, the last quarter of 2002). The firm fixed effects are included. The first-stage F-test statistics are reported. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is the last quarter of 2002 and the first quarter after May 2003. Only common stocks (those with a share code of 10 or 11 in the CRSP database) on NYSE are included in the sample. See Table 1 for other variable definitions.

	Algo		
	(1)	(2)	(3)
Post	0.021***	0.020***	0.022***
	(3.29)	(3.01)	(3.24)
Turnover	0.147***		
	(14.95)		
Volume		0.000***	
		(-5.02)	
Dollar Volume			0.000***
			(3.54)
ROA	0.209**	0.253**	0.217**
	(2.11)	(2.46)	(2.09)
Size	0.041***	-0.037***	-0.046***
	(12.20)	(-10.34)	(-12.23)
Leverage	0.000	0.003	0.015
	(0.01)	(0.20)	(1.04)
Analyst	-0.009**	-0.009**	-0.013***
	(-2.05)	(-2.05)	(-2.78)
Return	-0.066***	-0.068***	-0.073***
	(-4.98)	(-4.88)	(-5.25)
Dividend	-0.049	-0.037	-0.033
	(1.24)	(-0.82)	(-0.90)
SP500	-0.008	-0.006	-0.017*
	(-0.89)	(-0.63)	(-1.73)
Firm FE	Yes	Yes	Yes
F-statistics	18.95	19.37	20.16
N	1,618	1,618	1,618
adj. R^2	0.088	0.087	0.093

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Table 5. Effect of Algorithmic Trading on Firm Value using Fitted Value of Algorithmic Trading in the Two Quarters around Auto Quote

This table reports the results of the second stage regressions of the two-stage regression analysis for NYSE stocks in the two quarters surrounding the implementation of auto quote. The results of the first stage regressions are reported in Table 4. The fitted value of *Algo*, *Fitted Algo*, calculated from the first stage regressions is used in the second stage regressions of Tobin's q. Other independent variables in the second stage regression include *Spread*, *IVol*, *ISkew*, *Turnover* (model column (1)), *Volume* (column (2)), *Dollar Volume* (column (3)), *ROA*, *Size*, *Leverage*, *Analyst*, *Return*, *Dividend*, *SP500*, and firm fixed effect. T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is the last quarter of 2002 and the first quarter after May 2003. Only common stocks (those with a share code of 10 or 11 in the CRSP database) on NYSE are included in the sample. See Table 1 for other variable definitions.

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	Tobin's q								
	Panel A			Panel B			Panel C		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Fitted Algo	6.353***	6.738***	6.160***				2.046**	2.281**	2.014***
	(6.80)	(6.76)	(6.83)				(2.54)	(2.55)	(2.70)
Spread				-16.384***	-16.829***	-15.940***	-11.675***	-11.915***	-10.988***
				(-5.90)	(-5.82)	(-5.69)	(-4.38)	(-4.43)	(-4.02)
IVol				-11.277***	-11.237***	-11.190***	-9.912***	-9.881***	-9.734***
				(-4.44)	(-4.55)	(-4.82)	(-3.83)	(-3.90)	(-4.08)
ISkew				0.006*	0.005*	0.005*	0.006**	0.005*	0.006**
				(1.94)	(1.82)	(1.92)	(2.03)	(1.86)	(2.00)
Turnover	1.053***			0.114**			0.508***		
	(6.27)			(2.19)			(2.87)		
Volume		0.000***			0.000*			0.000***	
		(3.81)			(1.91)			(2.70)	
Dollar Volume			0.000**			0.000***			0.000***
			(2.25)			(3.14)			(2.80)
ROA	-1.279*	-1.731**	-1.347**	0.054	-0.017	0.001	-0.521	-0.748	-0.615
	(-1.87)	(-2.39)	(-1.97)	(0.09)	(-0.03)	(0.00)	(-0.79)	(-1.08)	(-0.93)
Size	-0.766***	-0.765***	-0.788***	-0.911***	-0.899***	-0.954***	-0.893***	-0.887***	-0.922***
	(-3.72)	(-3.67)	(-3.80)	(-4.12)	(-4.07)	(-4.27)	(-4.27)	(-4.20)	(-4.40)
Leverage	0.352	0.378	0.291	0.381	0.426	0.407	0.405	0.445	0.394
	(1.08)	(1.15)	(0.91)	(1.13)	(1.24)	(1.20)	(1.23)	(1.33)	(1.21)
Analyst	0.115***	0.197***	0.202***	-0.002	0.002	0.003	0.043	0.079**	0.087**
	(4.31)	(5.29)	(5.38)	(-0.14)	(0.11)	(0.16)	(1.55)	(2.03)	(2.20)
Return	0.605***	0.634***	0.634***	0.267***	0.264***	0.268***	0.407***	0.417***	0.429***
	(9.81)	(9.70)	(10.01)	(6.88)	(7.04)	(7.04)	(6.30)	(6.14)	(6.41)
Dividend	-0.250***	-0.392***	-0.379***	0.067**	0.064**	0.074***	-0.070	-0.130	-0.131*
	(-4.55)	(-5.30)	(-5.26)	(2.43)	(2.33)	(2.76)	(-1.19)	(-1.64)	(-1.68)
SP500	0.104	0.099	0.149*	0.061	0.068	0.056	0.055	0.057	0.073
	(1.20)	(1.12)	(1.68)	(0.55)	(0.60)	(0.49)	(0.52)	(0.52)	(0.67)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1,618	1,618	1,618	1,616	1,616	1,616	1,616	1,616	1,616
adj. R^2	0.174	0.172	0.207	0.209	0.207	0.237	0.215	0.213	0.245

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Table 6. Effect of Fitted Value of Algorithmic Trading, on Firm Value in the Two Quarters around Auto Quote Controlling Liquidity, Idiosyncratic Volatility and Idiosyncratic Skewness

This table reports the results of the second stage regressions of the two-stage regression analysis for NYSE stocks in the two quarters surrounding the auto quote. The results of the first stage regressions are reported in Table 4. The fitted value of *Algo*, *Fitted Algo*, calculated from the first stage regressions is used in the second stage regressions of Tobin's q. Other independent variables in the second stage regression include *Spread*, *IVol*, *ISkew*, *Turnover*, *ROA*, *Size*, *Leverage*, *Analyst*, *Return*, *Dividend*, *SP500*, and firm fixed effect. T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is the last quarter of 2002 and the first quarter after May 2003. Only common stocks (those with a share code of 10 or 11 in the CRSP database) on NYSE are included in the sample. See Table 1 for other variable definitions.

	Tobin's q				
	(1)	(2)	(3)	(4)	(5)
Fitted Algo	6.353***	3.276***	4.621***	5.957***	2.046**
	(6.80)	(3.20)	(4.61)	(6.71)	(2.54)
Spread		-15.388***			-11.675***
		(-6.08)			(-4.38)
IVol			-13.211***		-9.912***
			(-5.26)		(-3.83)
ISkew				0.009***	0.006**
				(3.00)	(2.03)
Turnover	1.053***	0.577***	0.809***	1.054***	0.508***
	(6.27)	(3.31)	(4.58)	(6.27)	(2.87)
ROA	-1.279*	-0.663	-0.924	-1.244*	-0.521
	(-1.87)	(-1.01)	(-1.36)	(-1.80)	(-0.79)
Size	-0.766***	-0.854***	-0.831***	-0.782***	-0.893***
	(-3.72)	(-4.07)	(-4.02)	(-3.80)	(-4.27)
Leverage	0.352	0.407	0.381	0.339	0.405
	(1.08)	(1.22)	(1.19)	(1.04)	(1.23)
Analyst	0.115***	0.057**	0.080***	0.114***	0.043
	(4.31)	(2.02)	(2.96)	(4.27)	(1.55)
Return	0.605***	0.437***	0.560***	0.551***	0.407***
	(9.81)	(6.86)	(8.92)	(8.74)	(6.30)
Dividend	-0.250***	-0.103*	-0.164***	-0.244***	-0.070
	(-4.55)	(-1.76)	(-2.87)	(-4.44)	(-1.19)
SP500	0.104	0.071	0.087	0.088	0.055
	(1.20)	(0.67)	(0.91)	(1.03)	(0.52)
Firm FE	Yes	Yes	Yes	Yes	Yes
N	1,618	1,618	1,618	1,616	1,616
adj. R^2	0.174	0.200	0.199	0.179	0.215

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Table 7. Effect of Algorithmic Trading on Firm Value Conditional on Liquidity, Firm Size and Implementation of Reg NMS

This table reports the effect of algorithmic trading on firm value conditional on effective spread, firm size, and implementation of Reg NMS. The dependent variable is Tobin's q. *Hispread* is equal to one for firms with effective spread higher than the industry mean in the quarter, and zero otherwise. *Large* is equal to one for firms with size greater than the industry mean in the quarter, and zero otherwise. *After07* is equal to one for the sub period after implementation of Reg NMS in 2007 (2008 to 2013) and zero for the sub period before the 2007 Reg NMS (2002 to 2006). T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is from 2002 through 2013. Only common stocks (those with a share code of 10 or 11 in the CRSP database) are included in the sample. See Table 1 for variable definitions.

	Tobin's q		
	(1)	(2)	(3)
Algo	0.454***	0.004	0.006
	(7.29)	(0.05)	(0.10)
Algo × Hispread	-0.412***		
	(-6.23)		
Algo × Large		0.497***	
		(5.85)	
Algo× After07			0.514***
			(5.35)
Hispread	-0.638***		
	(-16.87)		
Large		0.365***	
		(8.01)	
After07			0.457***
			(6.75)
Turnover	-0.000***	-0.000*	-0.000**
	(-3.02)	(-1.90)	(-2.07)
ROA	-0.690*	-0.685*	-0.690*
	(-1.71)	(-1.69)	(-1.71)
Size	-0.733***	-0.680***	-0.651***
	(-18.96)	(-16.79)	(-16.88)
Leverage	0.481**	0.413*	0.430**
	(2.21)	(1.90)	(1.96)
Analyst	0.233***	0.258***	0.264***
	(16.01)	(16.86)	(17.13)
Return	0.860***	0.824***	0.824***
	(25.33)	(24.84)	(24.78)
Dividend	0.055*	0.089***	0.096***
	(1.94)	(2.99)	(3.23)
SP500	0.389***	0.384***	0.388***
	(4.76)	(4.66)	(4.72)
Firm FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes
Observations	114,005	114,005	104,562
Adjusted R^2	0.179	0.165	0.163

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Table 8. Effect of Algorithmic Trading on Firm Value Conditional on Liquidity and Firm Size using Fitted Value of Algorithmic Trading in the Two Quarters around Auto Quote

This table reports the effect of algorithmic trading on firm value conditional on liquidity and firm size, using the fitted value of algorithmic trading for NYSE stocks in the two quarters surrounding the auto quote. Fitted value of algorithmic trading, *Fitted Algo*, is estimated in the first stage, where algorithmic trading (*Algo*) is regressed on a dummy variable, *Post*, and the control variable as in the second stage regressions. *Post* is set to one for the quarter after the auto quote (i.e., the first quarter after May 2003) and zero for the quarter before the auto quote (i.e, the last quarter of 2002). The fitted value of *Algo*, *Fitted Algo*, is used in the second stage regressions of Tobin's q. *Hispread* is equal to one for firms with effective spread higher than the industry mean in the quarter, and zero otherwise. *Large* is equal to one for firms with size greater than the industry mean in the quarter, and zero otherwise. T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is the last quarter of 2002 and the first quarter after May 2003. Only common stocks (those with a share code of 10 or 11 in the CRSP database) on NYSE are included in the sample. See Table 1 for other variable definitions.

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	Tobin's q					
	(1)	(2)	(3)	(4)	(5)	(6)
Fitted Algo	5.626***	5.638***	4.525***	4.429***	4.432***	3.517***
	(4.55)	(4.52)	(4.19)	(3.74)	(3.71)	(3.30)
Fitted Algo × Hispread	-0.641**	-0.668**	-0.647**			
	(-2.02)	(-2.05)	(-2.04)			
Hispread	-0.395**	-0.409**	-0.390**			
	(-2.21)	(-2.24)	(-2.18)			
Fitted Algo × Large				1.084**	1.087**	0.815**
				(2.21)	(2.23)	(1.98)
Large				0.629**	0.638**	0.496**
				(2.23)	(2.26)	(1.99)
Spread	-4.220**	-4.485**	-6.863**	-5.234***	-5.454***	-7.621***
	(-2.09)	(-2.16)	(-2.32)	(-2.66)	(-2.61)	(-2.66)
Turnover	0.072			0.072		
	(1.29)			(1.34)		
Volumel		0.000			0.000	
		(0.58)			(0.56)	
Dollar Volume			0.000**			0.000**
			(2.48)			(2.48)
ROA	-0.033	-0.075	-0.056	-0.047	-0.089	-0.061
	(-0.06)	(-0.13)	(-0.10)	(-0.08)	(-0.15)	(-0.11)
Size	-0.829***	-0.819***	-0.867***	-0.849***	-0.840***	-0.885***
	(-3.93)	(-3.89)	(-4.06)	(-3.97)	(-3.94)	(-4.09)
Leverage	0.428	0.454	0.430	0.409	0.433	0.408
	(1.29)	(1.35)	(1.27)	(1.24)	(1.30)	(1.22)
Analyst	0.201***	0.204***	0.163***	0.188***	0.191***	0.153***
	(4.18)	(4.14)	(3.57)	(3.98)	(3.94)	(3.39)
Return	0.645***	0.641***	0.561***	0.624***	0.622***	0.547***
	(6.73)	(6.71)	(6.91)	(6.58)	(6.56)	(6.77)
Dividend	0.060**	0.057**	0.068**	0.059**	0.057**	0.067**
	(2.12)	(2.06)	(2.48)	(2.08)	(2.01)	(2.41)
SP500	0.041	0.046	0.051	0.012	0.016	0.026
	(0.41)	(0.45)	(0.50)	(0.12)	(0.15)	(0.24)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,618	1,618	1,618	1,618	1,618	1,618
Adjusted R^2	0.217	0.216	0.232	0.214	0.212	0.228

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Table 9. Robustness Check: Lagged Algorithmic Trading and an Alternative Proxy of Algorithmic Trading

The table reports regression results of the effect of lagged algorithmic trading and an alternative proxy of algorithmic trading on firm value. The dependent variable is Tobin's q. The main independent variable is lagged algorithmic trading proxy, *Lagged Algo*. in Panel A, and the quote-to-trade (QT) ratio based algorithmic trading proxy, Algo_QT, in Panel B. See Table I for definitions of other variables. The industry classifications are defined by Fama and French (1997). Coefficients on firm, industry, and year-quarter dummies are not reported. T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is from 2002 through 2013. Only common stocks (those with a share code of 10 or 11 in the CRSP database) are included in the sample. See Table 1 for variable definitions.

	Tobin's q					
	Panel A			Panel B		
	(1)	(2)	(3)	(1)	(2)	(3)
Lagged Algo	0.307***	0.382***	0.370***			
	(5.59)	(6.79)	(6.33)			
Algo_QT				0.030***	0.034***	0.036***
				(6.39)	(7.17)	(7.31)
Turnover	-0.000**			-0.000**		
	(-2.04)			(-2.07)		
Volume		0.000***			0.000***	
		(6.01)			(6.11)	
Dollar Volume			0.000***			0.000***
			(3.19)			(3.28)
ROA	-0.780*	-0.771*	-0.779*	-0.697*	-0.687*	-0.695*
	(-1.84)	(-1.82)	(-1.83)	(-1.72)	(-1.70)	(-1.71)
Size	-0.649***	-0.659***	-0.669***	-0.648***	-0.659***	-0.669***
	(-16.84)	(-17.09)	(-17.56)	(-16.94)	(-17.21)	(-17.69)
Leverage	0.417*	0.421*	0.428*	0.427*	0.429**	0.439**
	(1.89)	(1.91)	(1.95)	(1.95)	(1.96)	(2.01)
Analyst	0.264***	0.261***	0.259***	0.262***	0.258***	0.255***
	(17.13)	(17.05)	(17.10)	(16.86)	(16.78)	(16.74)
Return	0.845***	0.844***	0.843***	0.827***	0.826***	0.826***
	(24.66)	(24.62)	(24.66)	(24.87)	(24.84)	(24.89)
Dividend	0.091***	0.103***	0.098***	0.096***	0.108***	0.103***
	(3.08)	(3.52)	(3.29)	(3.22)	(3.66)	(3.45)
SP500	0.390***	0.380***	0.377***	0.384***	0.375***	0.371***
	(4.72)	(4.64)	(4.78)	(4.63)	(4.55)	(4.69)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	112,486	112,487	112,487	113,992	113,993	113,993
Adjusted R^2	0.164	0.167	0.173	0.163	0.165	0.173

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Table 10. Effect of Change in Algorithmic Trading on Change in Firm Value

The table reports panel regression estimates of the effect of change in algorithmic trading on change in firm value. The dependent variable is change in Tobin's q, *Ch_Tobin's q*. The main independent variable is change in algorithmic trading proxy, *Ch_Algo*. All control variables are in the first differences. The industry classifications are defined by Fama and French (1997). T-statistics in parentheses are adjusted for both heteroskedasticity and within correlation clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is from 2002 through 2013. Only common stocks (those with a share code of 10 or 11 in the CRSP database) are included in the sample. See Table 1 for variable definitions.

	Ch_Tobin's q		
	(1)	(2)	(3)
Ch_Algo	0.243***	0.246***	0.258***
	(4.93)	(5.00)	(5.21)
Ch_Spread	-18.518***	-18.514***	-18.524***
	(-8.11)	(-8.11)	(-8.11)
Ch_Turnover	-0.000**		
	(-2.30)		
Ch_Volume		0.000	
		(0.66)	
Ch_Dollar Volume			0.000***
			(3.21)
Ch_ROA	-0.578**	-0.578**	-0.579**
	(-2.01)	(-2.01)	(-2.01)
Ch_Size	-1.891***	-1.892***	-1.897***
	(-18.85)	(-18.85)	(-18.91)
Ch_Leverage	1.447*	1.446*	1.446*
	(1.87)	(1.87)	(1.87)
Ch_Analyst	0.038***	0.038***	0.038***
	(4.38)	(4.36)	(4.37)
Ch_Return	0.705***	0.705***	0.706***
	(29.25)	(29.26)	(29.26)
Ch_Dividend	0.008	0.007	0.007
	(0.69)	(0.66)	(0.65)
Ch_SP500	0.089**	0.089**	0.078*
	(2.38)	(2.38)	(1.89)
Year-quarter FE	Yes	Yes	Yes
Observations	108,976	108,977	108,977
Adjusted R^2	0.257	0.257	0.259

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